50.055 Machine Learning Operations

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Important. This course handout serves as the single source of reference for *anything* that is related to this course. All updates will be reflected in this document.

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About this Course

In this module, students will learn the concepts and practice of machine learning operations ("MLOps"). The course does not teach the fundamentals of machine learning. Instead it focuses on building machine learning products. Students will learn how to identify viable use cases for machine learning, how to develop, test, deploy and monitor machine learning applications, how to manage machine learning projects and how to identify legal, ethical and data protection risks. Students will get hands-on experience on the complete machine learning development process, from experiments to deployment through course assignments and projects. The focus will be on deep learning and generative AI.
The course is offered to both undergraduate and graduate students as part of the regular SUTD curriculum. Alternatively (or additionally), it could be offered through SUTD Academy (CET).
As SUTD policy requires a full-time faculty to anchor each course, **an additional regular faculty member needs to be identified to co-teach the course.**

Learning Objectives

1. Identify good machine learning use cases
2. Conduct ML experiments and and train ML models
3. Prompt and finetune foundation models
4. Deploy models to production
5. Monitor and continuously improve models
6. Manage ML projects
7. Identify legal and ethical issues

Measurable Outcomes

1. Students can identify viable use cases
2. Students can conduct ML experiments
3. Students know how to prompt and finetune foundation models
4. Students understand ML architect and application stack
5. Students can create pipelines and deploy models
6. Students are able to conduct ethics, legal and data protection assessments

Assessment

Individual Assignments - One in Week 3 - One in Week 5	**20%** 10% 10%	Grade based on ***individual*** performance
Project / Group Assignments - Details to be released in Week 3 - Interim milestone in Week 6 - Final submission in Week 13	**20%** 10% 10%	Grade based on ***team*** performance
Class Participation - In-class activity, discussion	**10%**	Grade based on ***individual*** performance

Exams - Mid-term exam in week 8 - Final exam in week 14	**50%** 20% 30%	Grade based on ***individual*** performance

Course Prerequisites

50.007 Machine learning
50.039 Theory and Practice of Deep Learning or 60.001 Applied Deep Learning (can be co-requisite)

Or equivalent course on machine learning and deep learning

- 50.035 Computer Vision is accepted as replacement for 50.039
- 50.040 Natural Language Processing is accepted as replacement for 50.039
- 50.021 Artificial Intelligence is not a replacement for 50.039
- 50.045 Information Retrieval is not a replacement for 50.039

Course Policies

Academic Honesty

We take plagiarism seriously and we will take strict measures against students who plagiarise the work of others. If it is found that you have plagiarised in your submissions, all parties involved will receive an immediate zero for that assignment for your first offence. If you unwisely choose to repeat this, you will immediately fail the course. If students use generative AI tools, such as ChatGPT, they need to be transparent about its use and mark generated content as such.

Medical Certificates

- If you have a valid MC covering the date of the assessment, as per the norm, you must apply for an LOA with the university.
- We will decide on an appropriate means of grading you.
- Without an MC and LOA from the university, you will receive zero for the assessment.
- We will only accept one short term MC in the duration of the course. If you produce a second MC for a second assessment, **you will receive a zero for the assessment**. We will only accept a second MC if it pertains to you being hospitalised/quarantined/seriously ill.

Late Submission Policy

A late submission penalty will be applied to late submission (assignments or group assignment) as follows:
Between 0 and 6 hours late: -25%
6 to 12 hours late: -50%

12 to 18 hours late: -75%
Above 18 hours late: -100%

Instructor Office Hours

On request. Arrange appointment via email to

Daniel Dahlmeier (daniel_dahlmeier@sutd.edu.sg)
Christopher Teo (christopher_teo@sutd.edu.sg)

2026 Spring TA Contact: **Mark Huang** (he_huang@mymail.sutd.edu.sg) (Telegram)

Course Schedule & Deliverables

The course is planned for term 8 from January to April 2026. The course will run for 14 weeks. Week 1 to week 13 will be done physically, in class. There will be 3 sessions per week, 2 lectures and one hands-on tutorial for a total of 5 content hours per week. In addition, the students are expected to spend about 7 hours per week on self-study work for a total of 12 hours effort per week.

The detailed (tentative) schedule can be found below.

Week 1 - 6	26 January 2026 - 08 March 2026
Week 7 (Recess)	09 March 2026 - 15 March 2026
Mid-Term Examinations	20 March 2026 (Fr 11:30am - 1:30pm)
Week 8 - 14	16 March 2026 - 26 April 2026
Final Examinations	30 April 2026 (Thurs 1-3pm)

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Lectures/Activities

	Lecture (Mon 12-1330hrs) 1.5 hours Think Tank 2, 3	**Lecture (Tue 0900-1030hrs) 1.5 hours Think Tank 2, 3**	**Hands-on (Fri 1130-1330hrs) 2 hours IDIA Lab 2.605**
Week 1 (26 Jan)	1: Introduction(Chris)	2: Data	**Lab Work:** Train linear model and BERT classifier
Week 2 (2 Feb)	2: Data	3: ML experiments	**Lab Work:** Accelerate inference
Week 3 (9 Feb)	3: ML experiments	4: Foundation models	Group Assignment Brief | Q&A
Week 4 (16 Feb)	4: Foundation models	(Public holiday Chinese New Year)	**Lab Work:** Setup AWS and label data
Week 5 (23 Feb)	5: Foundation model fine-tuning	5: Foundation model fine-tuning	**Lab Work:** Train BERT classifier in AWS Sagemaker
Week 6 (2 March)	6: ML testing	6: ML testing	**Lab Work:** Prompt engineering with Gemini and Amazon Bedrock
Week 7 (9 Mar)	Recess		
Week 8 (16 Mar)	7:Deployment (Daniel, online)	7: Deployment (Daniel, online)	Mid-term exam (Chris) 20 March Think Tank 7, 8
Week 9 (23 Mar)	8: Monitoring and continuous improvement (Daniel)	8: Monitoring and continuous improvement (Daniel)	**Lab Work:** Retrieval Augmented Generation LLM-as-a-judge (Daniel)
Week 10 (30 Mar)	9: Project and change management (Daniel)	9: Project and change management (Daniel)	Public Holiday: Good Friday
Week 11 (6 Apr)	10: ML business cases and commercials(Daniel)	11: Legal and data protection (Daniel)	**Lab Work:** Finetune LLM model (Chris)
Week 12 (13 Apr)	12: AI ethics and regulation (Daniel)	**Lab Work:** Deploy LLM In Think Tank 2, 3 (Daniel)	(Capstone showcases) 17-18 Apr

Week 13 (20 Apr)	13: AI user experience(Daniel)	Guest Lecture(Goh Han Leong, SCS)	**Group Project Presentation**
Week 14 (27 Apr)	Final Exam (Daniel, Mark): 30 Apr 2026, Thursday, 5:00pm-7:00pm		

Public Holidays 2026 and other dates to note:

- Chinese new year 17-18 Jan 2026 (Tue-Wed)
- Hari raya 21 March 2026 (Sat)
- Good Friday 3 April (Fri)
- Labour Day 1 May (Fri)
- Capstone showcase Friday, 17 April (10am to 4pm) and Saturday, 18 April (10am to 3pm).

Lessons

Lectures

1. Introduction
    a. Recap of fundamental ML concepts
    b. ML products use cases
    c. When to use and when not to use ML
2. Data
    a. Types of data (structured, unstructured)
    b. Data-driven ML
    c. Data management and versioning
    d. Data cards
3. ML experiments
    a. ML development lifecycle
    b. Conduct and evaluate ML experiments
    c. Experiment tracking
4. Foundation models
    a. Foundation models and large language models
    b. Prompt engineering
    c. Retrieval-augmented generation
    d. Memory, chains and agents
5. Foundation model fine-tuning
    a. Parameter-efficient fine-tuning
    b. Instruction tuning
6. ML testing
    a. Data and model tests
    b. Evaluation metrics
    c. Foundation model evaluation
    d. Testing for bias and safety
    e. Creating and managing test sets
    f. Model cards

7. Deployment
    a. Deployment pipelines
    b. ML serving architectures
    c. Foundation model deployment
8. Monitoring and continuous improvement
    a. Logging and monitoring
    b. Data drift
    c. Retraining and improvement
9. Project and change management
    a. Managing ML teams
    b. Manage ML projects
    c. AI transformations
10. ML business cases and commercials
    a. ML companies and business models
    b. Pricing ML applications
11. Legal and data protection
    a. Legal basis for using data
    b. Data and model licences
    c. Data protection
12. AI ethics and regulation
    a. AI regulatory frameworks
    b. Ethics and bias
13. AI user experience
    a. AI design guidelines
    b. Conversational interfaces

Hands-on sessions

1. Introduction to AWS
    a. Overview of AWS platform and tools used in this course
    b. Setup environment
2. Data management and labelling
    a. Data storage on S3
    b. Labelling of data
    c. Check quality of annotations
    d. Version data sets
    e. Write data cards
3. Experiment tracking
    a. Perform ML experiments in Jupyter notebook
    b. Track experiments, configuration and metrics in Weights and Biases
4. Prompt engineering and retrieval-augmented generation
    a. Perform zero-shot and few-shot prompting
    b. Implement question answering bot with retrieval and answer generation capabilities
5. Fine-tuning and instruction tuning
    a. Implement parameter-efficient fine-tuning with small foundation model
    b. Implement instruction tuning
6. ML Testing
    a. Evaluate traditional ML tasks

   b. Evaluate quality and safety of foundation models
   c. Test for non-functional requirements
7. Deployment and serving
   a. Serve deep learning models for inference
   b. Deploy foundation models
8. Logging and monitoring
   a. Log technical and ML metrics
   b. Monitor performance

Readings

1. Chip Huyen. Designing Machine Learning Systems. 2022. O'Reilly
2. Kenji Lefevre. Introducing ML Ops. 2020. O'Reilly
3. Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. 2016. MIT Press
4. Additional academic papers from ACL, EMNLP, arxiv etc.
5. Technical blogs
6. Selected content from open-source online course, e.g., Full Stack Deep Learning course

Platforms and Tools

1. Python as programming language
2. Jupyter notebooks as main development environment
3. Pytorch as deep learning framework
4. AWS Sagemaker as general ML and compute platform
5. AWS S3 as object storage
6. Github for code version control
7. DVC for data versioning
8. Label studio for data annotation
9. Weights and Biases for experiment tracking
10. AWS Bedrock and Google AI studio for foundation model APIs
11. Langchain as a tool for model chains and agents
12. Pinecone as vector database
13. Huggingface as open-source foundation model platform

Access to SUTD GPU Cluster

For students enrolled in this course, you account in the SUTD GPU Cluster has been created, please note the following key information when accessing the cluster:

- Access to the cluster
  - **URL**: http://login2.gpucluster.sutd.edu.sg:8888
  - Alternative URL: http://192.168.33.15:8888/
  - You need to be connected to SUTD Network (either thru WiFi or VPN) and you need to use the default DNS server.
- You account:
  - Default **Username**: 100xxxx
  - Default **Password**: `AIcluster#2026`

- Each instance has a default wall time of 4 hours, meaning it will shut down automatically after 4 hours.
- Each user's home folder is persistent, so any new instance will load the same home storage. The default storage limit is 100 GB per user.

Course Assessment

To allow the students to gain practical experience with an end-to-end ML project, the course will emphasize practical data science and engineering tasks for assessment. The students will build a foundation model-based chatbot which can answer questions about SUTD to prospective students. The chatbot should be able to provide answers on a range of topics, e.g., available courses and degree programs, admission, campus life, and co-curricular activities.
The individual and group assignments will contribute building blocks to the chatbot, including data collection, prompt-engineering, information retrieval, evaluation, foundation model finetuning and deployment.
In addition, there will be a midterm and final exam testing the theoretical understanding of machine learning operations concepts.

Individual Assignments (20%)

1. Assignment 1: Data collection and prompt-engineering (10%)

This assignment requires students to create a data set for training and evaluation of an SUTD chatbot for prospective students. The data set should contain documents about SUTD and question-answer pairs suitable for model training and evaluation.
In addition to the data set, students should build a first prototype using only prompt engineering and foundation models available via APIs.
You will receive the instructions in **Week 3** and have until early **Week 5** to submit.

2. Assignment 2: Retrieval-augmented generation (10%)

This assignment will extend the chatbot prototype to include grounded answers. Instead of answering questions directly via a prompt alone, the bot should retrieve relevant documents and generate an answer based on the retrieved documents. The chatbot will continue to use models available via APIs. Students will be required to evaluate the performance of the chatbot.
You will receive the instructions in **Week 5** and have until early **Week 7** to submit.

Project / Group Assignment (20%)

For Group assignments, you will have to tackle a larger problem in scope.

Groups Creation

Groups will be created in Week 3 and will remain the same throughout the semester. Students will be free to create their own groups of 3 members.
Please fill in your group details at: [MLOps 2026 Student Grouping] before the end of Week 4.

Objective Overview

In this group project, you will design and build an LLM-based AI product or system around a clearly defined real-world problem. Your goal is to demonstrate that you can take an idea from problem definition → solution design → experimentation → evaluation → final delivery, using sound engineering judgment and quantitative evidence-driven decision making.

Project Requirements & Grading Rubrics

Please refer to the **[MLOps - Group Project Handout 2026]** for instructions and more information.

Midterm Exam (20%)

1. Date: 20/03/2026, Duration: 1 hour. Location: Think Tank 7, 8
2. Will be conducted **in person**, on **paper.**
3. Questions format will be multiple choice, short and long answers.
4. This will be a **closed book** exam.
5. You will be allowed to bring a cheat sheet with the following requirements:
    a. **One** A4 page (21x29.7 cm)
    b. Hand-written only
    c. Double-sided

Final Exam (30%)

6. Date:**30 Apr 2026 (Thursday)**, **5:00pm-7:00pm**. Duration: 2 hours. Location: **ISH 1**
7. Will be conducted **in person**, on **paper.**
8. Questions format will be multiple choice, short and long answers.
9. This will be a **closed book** exam.
10. You will be allowed to bring a cheat sheet with the following requirements:
    a. **One** A4 page (21x29.7 cm)
    b. Hand-written only
    c. Double-sided

Participation (10%)

Includes attendance and participation in class activities.

Auditing Students

1. Auditing students are not required to take the exam
2. Auditing students have to complete at least one personal assignment
3. Auditing students are thus only required to attend the lectures

Related ML courses at SUTD

- 50.006 User interface design and implementation. Focuses on UI/UX in general and not on UX for AI in particular

- 50.007 Machine learning. Focuses on machine learning fundamentals and not building ML products
- 50.038 Computational Data Science. Focuses on a complete data science project with visualisation, implementation and explanations. Does not focus on building ML products
- 50.039 Theory and Practice of Deep Learning. Focused on fundamentals of deep learning
- 60.001 Applied Deep Learning. Focuses on applying deep learning but not non-technical aspects like legal, data protection, business case
- 02.143DH Artificial Intelligence and ethics. Focuses more on philosophical questions and not on building ML products.